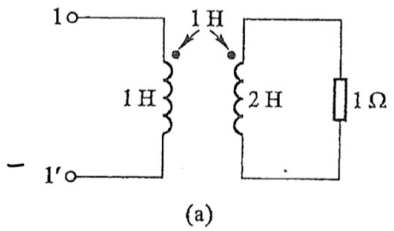
10-5 求题 10-5 图所示电路的输入阻抗 $Z(\omega=1\ \text{rad/s})$。

(a) $Z = Z_{11} + \dfrac{(\omega M)^2}{Z_{22}} = j\omega L_1 + \dfrac{1}{j\omega L_2 + 1}$

$= \dfrac{1}{5} + j\dfrac{3}{5}$



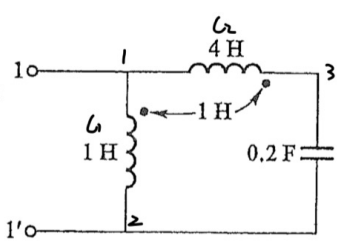
(a)

(b) 等效为：

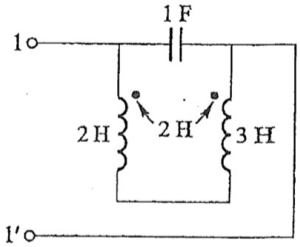


$Z = (j\omega(L_2+M) - \frac{1}{j\omega C}) \,//\, j\omega(L_1+M) - j\omega M$

$= -j1\ \Omega$



(b)



(c)

(c) $Z = -j\frac{1}{\omega C} \,//\, j\omega(L_1+L_2-2M)$

$= -j \,//\, j = \infty$

⇒ 等效为开路

题 10-5 图

10-6 题 10-6 图所示电路中，$R_1=R_2=1\ \Omega$，$\omega L_1=3\ \Omega$，$\omega L_2=2\ \Omega$，$\omega M=2\ \Omega$，$U_1=100\ \text{V}$。求：

(1) 开关 S 打开和闭合时的电流 $\dot{I}_1$。

(2) S 闭合时各部分的复功率。

(1). S打开：$Z = R_1 + R_2 + j\omega(L_1+L_2+2M) = 2+j9\ \Omega$

设 $\dot{U}_1 = U_1\angle 0°$，$\dot{I}_1 = \dfrac{\dot{U}_1}{Z} = \dfrac{40}{17} - j\dfrac{180}{17}\ \Omega$

S闭合：$Z = R_1 + j\omega L_1 + R_2$ ✗

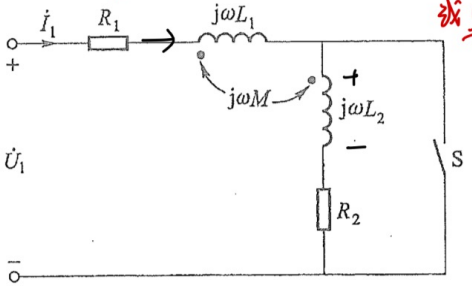
$\dot{U} = \dot{U}_1 - \dot{U}_{L2} = \dot{U}_1 - j\omega M\dot{I}_1$

$\dot{I}_1 = \dfrac{\dot{U}}{Z} = \dfrac{\dot{U}_1 - j\omega M\dot{I}_1}{R_1 + j\omega L_1 + R_2}$

$\Rightarrow \dot{I}_1 = \dfrac{200}{29} - j\dfrac{500}{29}\ \Omega$

$\bar{S} = \dot{U}\cdot\dot{I}_1^* = \dfrac{20000}{29} + j\dfrac{20000}{29}\ \text{V·A}$

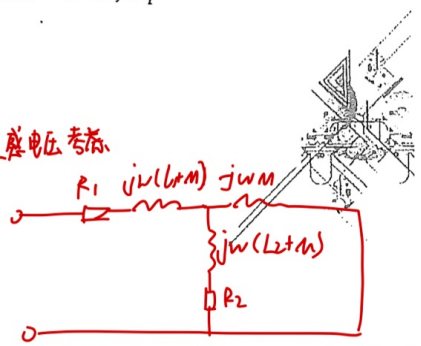
$\bar{S}_{L1} =$



题 10-6 图

改为KVL，互感电压考虑

戴等效：



$Z = R_1 + j\omega(L_1+M) + (j\omega(L_2+M)+R_2) \,//\, (-j\omega M)$

$\dot{I}_1 = \dfrac{\dot{U}_1}{Z} = 43.85\angle -37.88°\ \text{A}$

(2) KVL ⇒ $\dot{I}_1 = 43.85\angle -37.88°\ \text{A}$，$\dot{I}_2 = 39.22\angle 18.69°\ \text{A}$

$\dot{U}_1\dot{I}_1^* = (1+j3)\dot{I}_1\dot{I}_1^* + j2\dot{I}_2\dot{I}_1^*$

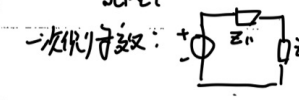
$j2\dot{I}_2\dot{I}_1^* + (1+j2)\dot{I}_2^2 \Rightarrow$

$(1922.08+j576.23)$ 为 $(R_1+j\omega L_2)$ 吸收，$1567.96 - j376.66$

10-12 题 10-12 图所示电路中 $R=1\ \Omega$，$\omega L_1=2\ \Omega$，$\omega L_2=32\ \Omega$，耦合因数 $k=1$，$\dfrac{1}{\omega C}=32\Omega$。求电流 $\dot{I}_1$ 和电压 $\dot{U}_2$。

$k = \dfrac{M}{\sqrt{L_1L_2}} \Rightarrow M = 8\ \Omega$　$Z_{11} = R + j\omega L_1 = 1 + j2\ \Omega$，$Z_{22} = \dfrac{1}{j\omega C} + j\omega L_2 = 0$

一次侧等效：



$Z_L = \dfrac{(\omega M)^2}{Z_{22}} = \infty$

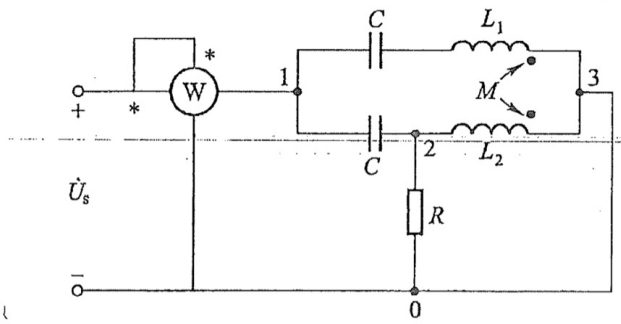
$\Rightarrow \dot{I}_1 = 0$

二次侧：$\because \dot{I}_1 = 0 \quad \therefore j\omega M\dot{I}_1 = 0$

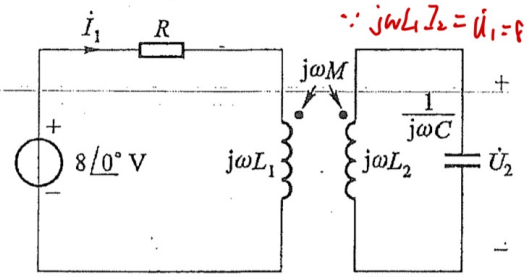
$\therefore \dot{U}_2 = 0$ ✗　$\dot{U}_{L2} + \dot{U}_{L2} = 0$ 但 $\dot{U}_2 \neq 0$

$\because j\omega L_1\dot{I}_2 = \dot{U}_1 = 8\angle 0°\ \text{V} \quad \therefore \dot{I}_2 = j1\ \text{A}$

$\therefore \dot{U}_2 = \dot{I}_2 \cdot \dfrac{1}{j\omega C} = 32\text{V}$

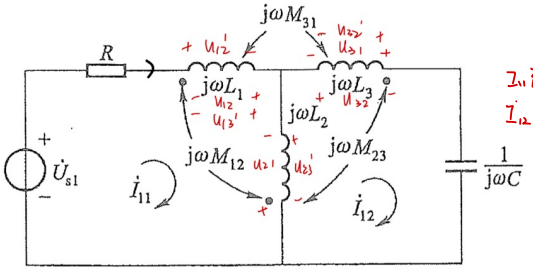


题 10-11 图



题 10-12 图

10-15 列出题 10-15 图所示电路的回路电流方程。



题 10-15 图

$\dot{I}_{11}$ 产生的互感电压：u_{21} / u_{31}　$u_{12}=u_{21}=j\omega M_{12}\dot{I}_{11}$，$u_{13}=j\omega M_{31}\dot{I}_{11}$，$u_{32}=j\omega M_{23}\dot{I}_{11}$

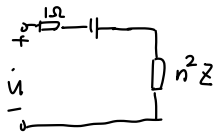
$\dot{I}_{12}$ 产生的互感电压（u_{12}'，u_{13}'）：$u_{12}'=j\omega M_{12}\dot{I}_{12}$，$u_{32}'=u_{23}'=j\omega M_{23}\dot{I}_{12}$，$u_{13}'=j\omega M_{13}\dot{I}_{12}$

$$\begin{cases}(R+j\omega L_1+j\omega L_2)\dot{I}_{11}-j\omega L_2\dot{I}_{12}-j\omega M_{31}\dot{I}_{11}-j\omega M_{12}\dot{I}_{11}-j\omega M_{23}(\dot{I}_{11}-\dot{I}_{12})+j\omega M_{23}(\dot{I}_{11}-\dot{I}_{12})=\dot{U}_{s1} \\ -j\omega L_2\dot{I}_{11}+(j\omega L_2+j\omega L_3+\frac{1}{j\omega C})\dot{I}_{12}=(j\omega M_{31}-j\omega M_{23})\dot{I}_{11}-(j\omega M_{12}-j\omega M_{23})(\dot{I}_{11}-\dot{I}_{12})\end{cases}$$

$$\begin{cases}(R+j\omega L_1+j\omega L_2)\dot{I}_{11}-j\omega L_2\dot{I}_{12}+u_{12}'-u_{12}-u_{13}'-u_{21}+u_{23}'=\dot{U}_{s1} \\ -j\omega L_2\dot{I}_{11}+(j\omega L_2+j\omega L_3+\frac{1}{j\omega C})\dot{I}_{12}-u_{23}'+u_{21}+u_{32}-u_{31}-u_{32}'=0\end{cases}$$

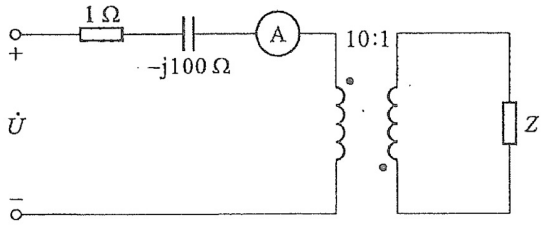
10-18 求题 10-18 图所示电路中的阻抗 Z。已知电流表的读数为 10 A，正弦电压有效值 $U=10$ V。

一次侧等效：



$$\frac{\dot{U}}{\dot{I}}=1-j100+100Z$$

$$\Rightarrow \frac{10}{10}=1-j100+100Z \Rightarrow Z=j1\ \Omega$$



题 10-18 图

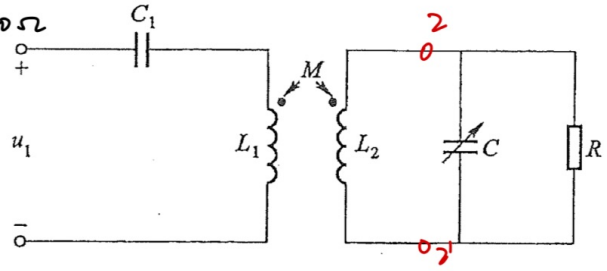
10-22 题 10-22 图所示电路中 $C_1=10^{-3}$ F，$L_1=0.3$ H，$L_2=0.6$ H，$M=0.2$ H，$R=10\Omega$，$u_1=100\sqrt{2}\cos(100t-30°)$ V，C 可变动。C 为何值时，R 可获得最大功率？并求出最大功率。

$\omega=100$ rad/s

$Z_{11}=-j\frac{1}{\omega C_1}+j\omega L_1=j20\ \Omega$

⇒ 二次侧等效：





题 10-22 图

$U_{oc}=j\omega M\cdot\frac{\dot{U}_1}{Z_{11}}=100\angle 30°$ V

$Z_i=\frac{(\omega M)^2}{Z_{11}}=-j20\ \Omega$

$Z_{eq}=Z_i\,//\,\frac{1}{j\omega C}=j\frac{1}{20C-\frac{1}{5}}\ \Omega$

$R=|Z_{eq}|$ 时功率最大 $\Rightarrow \frac{1}{20C-\frac{1}{5}}=10 \Rightarrow C=$

$Y_{eq}=-j0.025$ S $\Rightarrow \dot{I}_{sc}=2.5\angle -120°$ A



当 R 上电压最大时，功率最大

$Y_{eq}+j\omega C=0$，$\dot{I}_{sc}$ 全流过

$\omega C=0.025$　$C=250\,\mu$F

$P=I_{sc}^2R=62.5$ W.